

Study Guide - LIN 313: Language and Computers (Fall 2025)

Use these review questions along with past homeworks and exams and course materials to prepare for the final. This practice has more questions than will appear on the actual test.

Instructions for actual exam:

- Time limit: 2 hours (same as actual exam)
- One 8.5"x11" double-sided note sheet allowed
- No calculators or electronic devices
- Show all work for partial credit

1 True/False with Justification

For each statement, circle TRUE or FALSE. Then provide a brief justification (1-2 sentences).

1. The Word2Vec model can distinguish between "bank" (financial institution) and "bank" (river edge) based on the sentence context.

TRUE / FALSE

Justification: _____

2. In a transformer model, the attention mechanism allows each token to "see" every other token in the input sequence.

TRUE / FALSE

Justification: _____

3. RLHF completely eliminates the possibility of a language model producing harmful or false content.

TRUE / FALSE

Justification: _____

4. If two word embeddings have a high cosine similarity in vector space, they are guaranteed to be synonyms.

TRUE / FALSE

Justification: _____

5. The "distributional hypothesis" states that words which appear in similar contexts tend to have similar meanings.

TRUE / FALSE

Justification: _____

2 Multiple Choice

Circle the BEST answer for each question.

6. Which of the following is NOT a component of the standard RLHF pipeline?
 - (a) Pre-training on large text corpora
 - (b) Supervised fine-tuning on demonstration data
 - (c) Training a reward model on human preferences
 - (d) Manually editing the model's weights to encode facts
7. A researcher finds that in their word embedding space, $\text{vec}(\text{"doctor"}) - \text{vec}(\text{"man"}) + \text{vec}(\text{"woman"}) \approx \text{vec}(\text{"nurse"})$. This result most directly illustrates:
 - (e) The power of neural networks to learn medical terminology
 - (f) That word embeddings encode societal biases from training data
 - (g) The mathematical precision of vector arithmetic
 - (h) That doctors and nurses have similar job responsibilities
8. The "sycophancy" problem in RLHF-trained models refers to:
 - (i) Models that refuse to answer any questions
 - (j) Models that tell users what they want to hear rather than the truth
 - (k) Models that copy text verbatim from training data
 - (l) Models that generate text too slowly
9. A company deploys a customer service chatbot using retrieval-augmented generation (RAG). The system retrieves relevant passages from a product documentation database before generating responses. During testing, a user asks: "What is the warranty period for the X500 model?" The retrieved passage states: "The X500 comes with a standard 2-year warranty." The retrieved passage is appended to the user's prompt. However, the chatbot responds: "The X500 comes with our industry-leading 3-year comprehensive warranty, which covers all parts and labor." Which of the following best explains why RAG failed to prevent hallucination in this case?
 - (m) The retrieval system failed to find the relevant passage, forcing the model to rely on parametric knowledge.
 - (n) The retrieved passage was relevant, but the language model's generation process is not constrained to faithfully reproduce or reason over retrieved content.
 - (o) RAG systems can only prevent hallucinations about facts that appear verbatim in the retrieved documents, not paraphrased information. Models that generate text too slowly
 - (p) The embedding similarity between the query and passage was below the retrieval threshold, so the passage was filtered out before generation.
10. In the above scenario, why is the chatbot more likely to hallucinate "The X500 comes with our industry-leading 3-year comprehensive warranty, which covers all parts and labor" rather than "The X500 has no warranty"?
 - (q) The model's temperature setting was too high, causing it to generate longer and more elaborate responses.
 - (r) During RLHF training, human raters likely preferred responses that sounded helpful, confident, and customer-friendly, reinforcing patterns of optimistic, elaborated language over terse or negative statements.
 - (s) The X500 warranty information was included in the model's pretraining data more frequently than other products, causing the model to overfit to that specific warranty length.
 - (t) The retrieval system prioritizes passages with positive sentiment, filtering out any documentation about warranty limitations.

	Doc 1	Doc 2	Doc 3	Doc 4	Doc 5
the	24	30	18	19	28
of	16	12	14	14	10
gas	3	0	1	0	0
ticket	1	2	6	0	0
miles	0	1	2	0	0
suitcase	2	2	3	0	0
library	0	1	0	2	5
study	0	0	1	7	4
meet	2	1	0	0	3
procrastinate	0	0	0	1	8

Above is a Term X Document matrix constructed from a corpus of 5 documents. Documents 1-3 are about winter break, and while documents 4-5 are about finals week. The matrix contains raw co-occurrence counts, before any preprocessing.

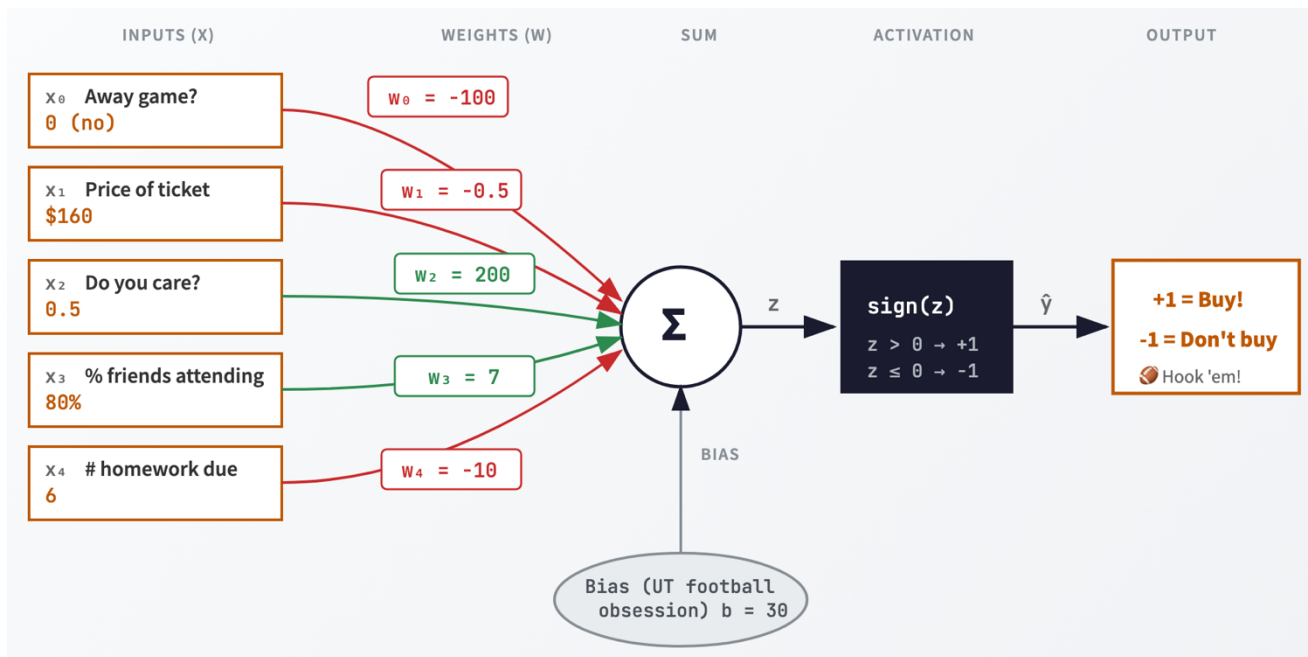
11. Using raw term frequencies, Document 4 (a "finals" document) has a cosine similarity of 0.94 with Document 1 (a "break" document), but only 0.92 with Document 5 (another "study" document). This is counterintuitive. Why would a study document be MORE similar to a travel document than to another study document?

- (u) The cosine similarity calculation contains a mathematical error when documents share zero-frequency terms.
- (v) Document 4 and Document 1 happen to have similar magnitudes for the high-frequency function words ("the," "of"), which dominate the dot product and obscure the topical differences encoded in lower-frequency content words.
- (w) Document 5 contains the word "procrastinate" 8 times, which is an outlier that artificially decreases its similarity with all other documents.
- (x) Cosine similarity only measures the angle between vectors, so documents with more total words will always appear more similar to each other.

12. (Short answer) Think of one data pre-processing step that would help fix the problem above (or describe one you know), so that finals documents "cluster" together and break documents "cluster" together. How would your measure help fix the issue?

3 Perceptron

Consider the following perceptron model



(a) Calculate the dot product of the input vector and the weight vector (for x_3 , use the percentage as a number rather than a decimal, so 80 instead of .8)

(b) Complete the rest of the forward pass. For these inputs, does the model say you should go to the game or not?

(c) All of the greatest minds in the machine learning research departments at UT have teamed up to train a powerful perceptron model to predict whether you should go to the UT game. To collect gold training data, before every game they had every student at UT record answers to all of the input “features” and note whether they attended or not. They unveil an app called “ask BEVO” designed to help all the world make this very important decision. Ever UT student’s opinions informed the model training. Why won’t all of the students be happy with the app’s predictions for them?

4 Attention Analysis

Below is a self-attention visualization for the beginning of a Winograd schema "The cat sat on the mat because it was....." There is one word missing. The row represents the query token "it"; columns represent the tokens in the rest of the input. Higher values indicate stronger attention.

	The	cat	sat	on	the	mat	because	it	was
it	.05	.28	.08	.02	.03	.72	.02	.01	.01

(a) Based on the attention pattern, what is a likely last word in the sentence? How can you tell from the attention weights?

(b) This type of pronoun resolution is an example of what NLP problem? Why is attention particularly helpful for solving it?

5 Conversational Pragmatics

For each chatbot response below, identify which Gricean maxim is violated and explain how.

Scenario A: User asks: "What's $2 + 2$?" Chatbot responds: "The answer is 5."

Maxim violated: _____ Explanation: _____

Scenario B: User asks: "Can you recommend a good Italian restaurant?" Chatbot responds: "The mitochondria is the powerhouse of the cell."

Maxim violated: _____ Explanation: _____

Scenario C: User asks: "Is this medication safe?" Chatbot responds: "Well, it depends on what you mean by 'safe' and also 'medication' has many definitions and the concept of 'is' raises philosophical questions about existence..."

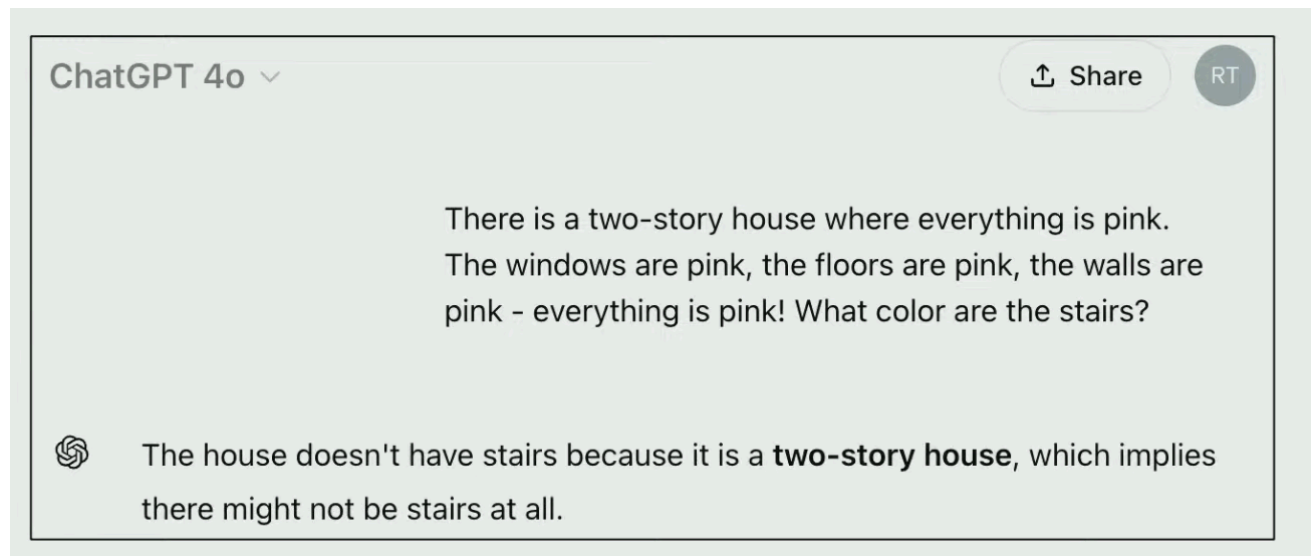
Maxim violated: _____ Explanation: _____

6 Model Comparisons

Fill in the comparison table for different language modeling approaches. The answer might not be immediately obvious for some of the models (for example, we didn't talk explicitly about N-gram models and polysemy). For unclear cases, reason about it as best you can and provide a justification. (You won't be asked a comprehensive question like this on the exam, but use this table to your organize your thinking about all of the machine learning methods we have covered and the relations between them).

Feature	Naïve Bayes	N-gram Model	Perceptron	Count-based distributional model	Word2Vec	Transformer (BERT, ChaptGPT)
Handles polysemy?						
Context window/input size: fixed or variable? What size?						
Captures semantics?						
Captures syntax?						
Dense vs. Sparse internal representations						
Supervised / Unsupervised / "Self-supervised"						

7 Short Answer Questions



(a) ChatGPT answers the above brainteaser incorrectly (though it gets trickier questions correct). Why does it answer in this way?

(b) The homework discussed "affordances" of words like "book" and "apple." Give two different affordances for the word "key."

(c) Why doesn't in-context learning (a.k.a few-shot or many-shot learning using LLMs, the way we trained the red-flag classifier) require as much training data as traditional machine learning classification methods?

8 Essay Question

Choose ONE of the following prompts. Write a well-organized response with a main argument, supporting points and illustrative examples (about 1 page).

Option A: When users "jailbreak" a chatbot, some claim they are revealing the model's "true self" hidden beneath corporate restrictions. Others argue there is no "true self" to reveal. Drawing on what you've learned about how LLMs work (training data, RLHF, system prompts, next-token prediction, personas, genres, registers), take a position: Does jailbreaking reveal a chatbot's authentic nature? Why or why not? To support your position, your answer should address what technically happens when a jailbreak succeeds.

Option B: The philosopher John Searle argued with his "Chinese Room" thought experiment that computers can never truly understand language—they just manipulate symbols. Now that we have LLMs that can write poetry, answer questions, and hold conversations, has Searle been proven wrong? Take a position and defend it with specific examples from what you've learned about how LLMs work.

Option C: You are advising a company that wants to deploy an LLM chatbot for customer service. Drawing on concepts from the course (training data bias, hallucination, RLHF, interpretability, Gricean maxims), write a memo outlining: (1) three specific risks they should be aware of, and (2) concrete steps to mitigate each risk.